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CS101C: Introduction to Programming (Using C)

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CSE Department

Indian Institute of Technology, Dharwad

Week 12

Today's Class (26/10/2026)

From `scanf()` to `fscanf()`



You already know how to read input from the keyboard!

From `scanf()` to `fscanf()`

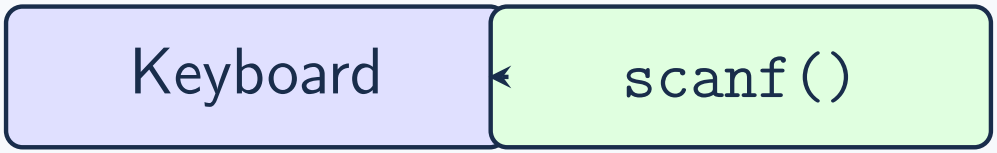
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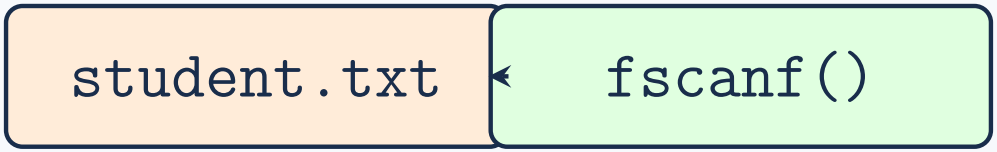
You already know how to read input from the keyboard!

Reading from the Keyboard

```
int age;
scanf("%d", &age);
```

- `scanf()` reads formatted input from `stdin`, usually the keyboard.
- But what if the information is already stored in a file?





Meet `fscanf()`!

It reads formatted data from a specified input stream, such as a file.

Syntax of `fscanf()`

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```
fscanf(fp, format, arguments...)
```

`fp`	The input stream associated with an opened file.
`format`	Specifies the expected data types, such as `%d`, `%s`, and `%f`.
`arguments`	Addresses of the variables where the input values will be stored.

Example

```
fscanf(fp, "%d %29s %f", &roll, name, &marks);
```

- `&roll`: store an integer in `roll`.
- `name`: store a string in the character array.
- `&marks`: store a floating-point value in `marks`.

Our Input File: `student.txt`

Suppose the file contains:

```
209 Aikta 100
```

The file contains three pieces of information:

Value	Meaning	Data type
209	Roll number	`int`
`Aikta`	Student name	String
100	Marks	`float`

Our Goal

Read these three values from the file and store them in three different variables.

Program: Reading Student Data

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```c
#include <stdio.h>
int main(void)
{
    FILE *fp = fopen("student.txt", "r");
    int roll;
    char name[30];
    float marks;
    if (fp == NULL)
        return 1;
    if (fscanf(fp, "%d %29s %f", &roll, name, &marks) == 3)
    {
        printf("Roll: %d\n", roll);
        printf("Name: %s\n", name);
        printf("Marks: %.1f\n", marks);
    }
    fclose(fp);
    return 0;
}
```

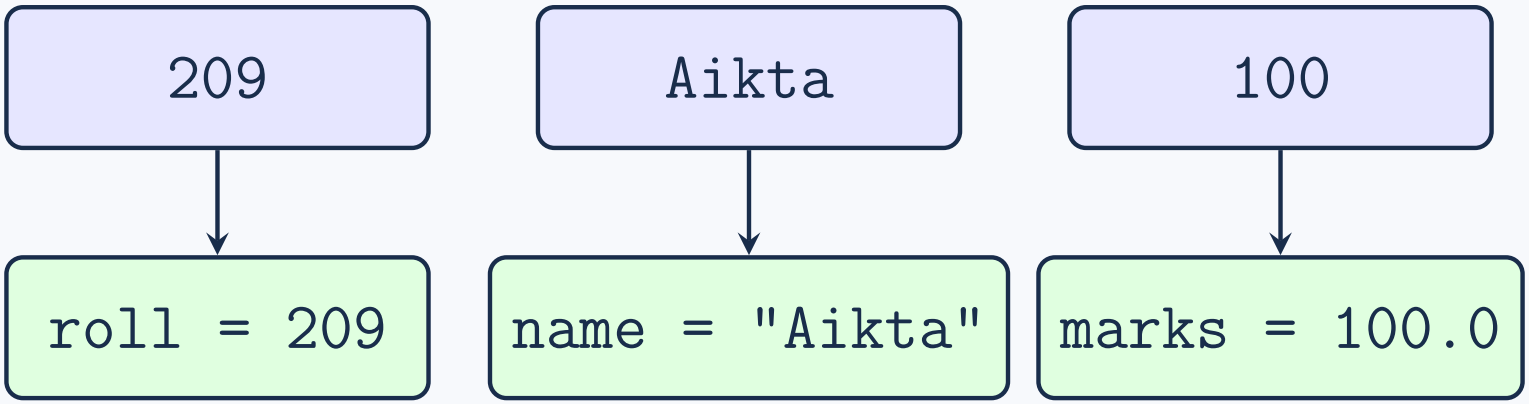
Always check whether `fopen()` succeeded. The condition `== 3` checks that all three input items were successfully assigned.

How Does the Data Get Stored?

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The statement is:

```
fscanf(fp, "%d %29s %f",
           &roll, name, &marks);
```



- `%d` reads the integer 209.
- `%29s` reads the word `Aikta`.
- `%f` reads the number 100.

Understanding the Output

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After reading the file, the variables contain:

Variable	Stored value
`roll`	`209`
`name`	`"Aikta"`
`marks`	`100.0`

The program prints:

```
Roll:  209
Name:  Aikta
Marks:  100
```

Notice: `printf()` displays the values on the console; `fscanf()` reads them from the file.

What Does `fscanf()` Return?

`fscanf()` returns the number of input items successfully matched and assigned.

File contents	Return value
`209 Aikta 100`	3
`209 Aikta`	2
`209 Aikta abc`	2

Why?

- First item: roll number.
- Second item: name.
- Third item: marks.

If the marks value is missing or cannot be read as a floating-point number, only the first two items may be assigned.

A return value of 3 means all three items were assigned successfully. It does not mean the file has exactly three items.

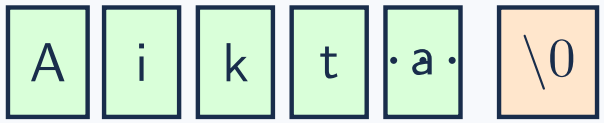
Why Do We Write %29s?

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We declared:

```
char name[30];
```

This array can store at most **29 characters of the name plus one terminating character** `'\0'`.



- %29s reads at most 29 characters.
- The remaining array space allows the terminating `'\0'`.
- Plain %s has no width limit and may overflow the array.

Another limitation

%s stops at whitespace. Therefore, it reads `Aikta` from `Aikta Arya`, not the full name.

Quick Challenge: Predict the Return Value

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Suppose `student.txt` contains:

```
205 Ravi abc
```

The program executes:

```
int roll;
char name[30];
float marks;

int result = fscanf(fp, "%d %29s %f",
                    &roll, name, &marks);
```

What is the value of `result`?

1. 0
2. 1
3. 2
4. 3

Answer

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`result = 2`

1. `%d` successfully reads `205`.
2. `%29s` successfully reads `Ravi`.
3. `%f` cannot convert `abc` into a floating-point value.

Therefore, two input items are successfully assigned.

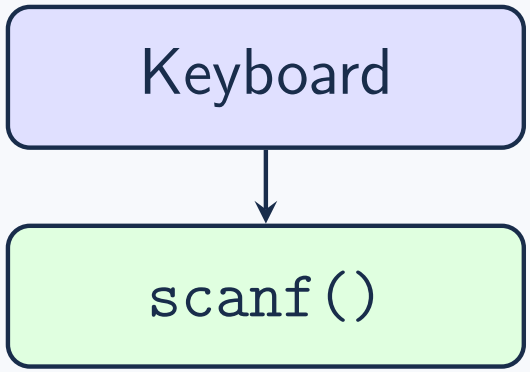
Remember

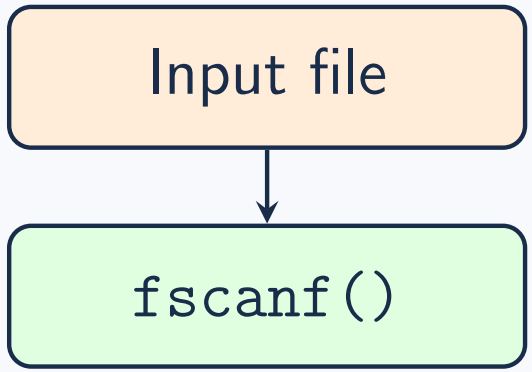
The failed conversion does not erase the values already read. Also, the unmatched input may remain in the stream.

`scanf()` vs `fscanf()`

Both functions read formatted input. The main difference is **where the input comes from**.

Function	**Input source**
`scanf("%d", &x)`	Standard input (`stdin`), usually the keyboard
`fscanf(fp, "%d", &x)`	The stream `fp`, such as an opened file





What is common?

Both use format specifiers such as %d, %f, %c, and %s.

Where Do We Use the & Symbol?

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Consider these variables:

```
int age;
char name[30];
```

To read them from a file:

```
fscanf(fp, "%d %29s", &age, name);
```

Argument	Reason
`&age`	Passes the address of the integer variable
`name`	Passes the array's starting address

Remember

For ordinary variables, pass their addresses using &. For a character array used with %s, pass the array name directly and specify a width limit.

The Format Specifier Trap!

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Suppose `data.txt` contains:

```
25 3.14 Z
```

Our variables are:

```
int a;
float b;
char c;
```

Now consider:

```
fscanf(fp, "%d %f %c", &a, &b, &c);
```

Predict the values

- `%d` reads the integer.
- `%f` reads the floating-point value.
- `%c` reads one character.

Trace the Input Step by Step

The file initially contains:

```
25    3.14    Z
```

Specifier	Value assigned	Remaining input
%d	a = 25	3.14 Z
%f	b = 3.14	Z
%c	c = 'Z'	Empty

Therefore:

$$a = 25, \qquad b = 3.14, \qquad c = \text{'Z'}$$

Important detail

Whitespace in the format string skips leading whitespace for most conversions, including %d and %f. But %c does not skip leading whitespace on its own.

Now Remove %c from the Format!

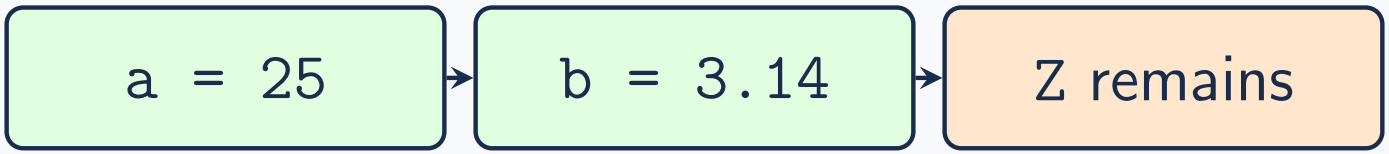
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What if we write:

```
fscanf(fp, "%d %f", &a, &b);
```

The file still contains:

```
25   3.14   Z
```



- Two input items are assigned successfully.
- The return value is 2.
- The character `Z` is not consumed by this call.

Key lesson

The format string determines which input conversions are attempted. Unrequested input can remain in the stream.

Quick Challenge

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Suppose `data.txt` contains:

```
25 3.14 Z
```

Predict the result of:

```
int a;
float b;
char c;
fscanf(fp, "%d %f", &a, &b);
fscanf(fp, "%c", &c);
```

What is the value of `c`?

1. `'Z'`
2. `' '`
3. `'3'`
4. The program must fail

Quick Challenge

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Suppose `data.txt` contains:

```
25 3.14 Z
```

Predict the result of:

```
int a;
float b;
char c;
fscanf(fp, "%d %f", &a, &b);
fscanf(fp, "%c", &c);
```

What is the value of `c`?

1. `'Z'`
2. `' '`
3. `'3'`
4. The program must fail

Hint: Does %c skip whitespace automatically?

Challenge Answer: A Whitespace Trap! (Last)

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Option 2: $c =$ ' '

Why?

- The first call reads 25 and 3.14.
- The whitespace between 3.14 and Z remains in the stream.
- The second call uses %c, which reads the next character exactly as it is, including whitespace.

To skip leading whitespace before reading the character:

```
fscanf(fp, " %c", &c);
```

Remember

The space before %c in the format string tells `fscanf()` to skip any leading whitespace. Then c receives 'Z'.

Today's Class (28/10/2026)

From `printf()` to `fprintf()`

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You already know how to display formatted output!

Displaying on the Screen

```
int age = 20;
printf("Age = %d\n", age);
```

Output on the console:

```
Age = 20
```

But what if we want to save the formatted output in a file instead?

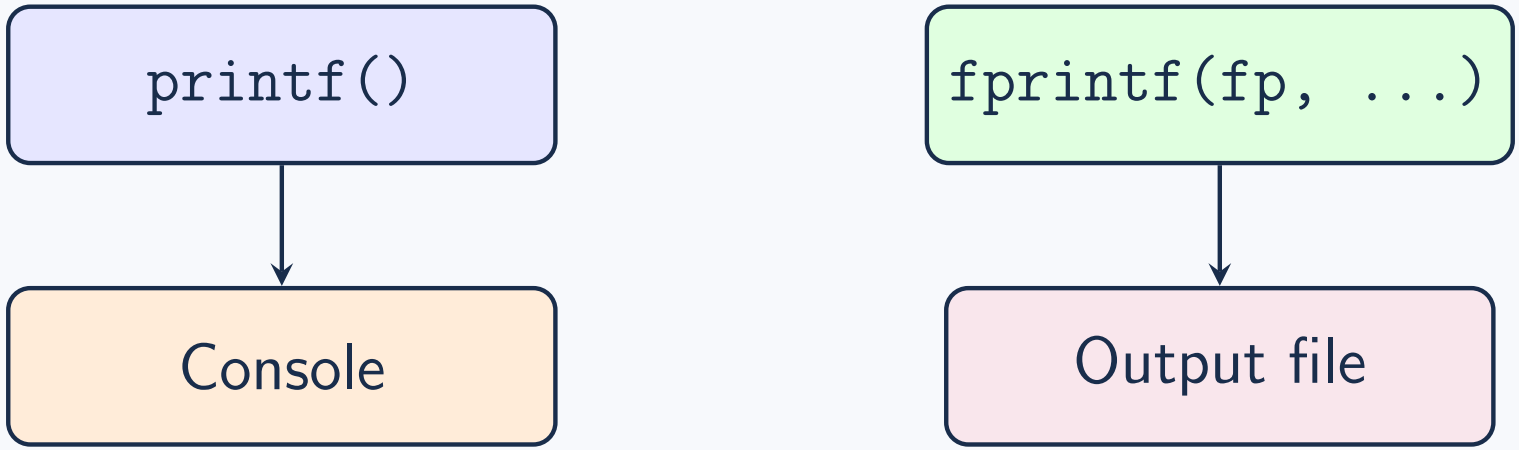
Meet `fprintf()`!

It writes formatted output to a specified stream, such as a file.

Where Does the Output Go?

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The key difference is the **stream argument**.



- `printf()` writes to standard output, usually the console.
- `fprintf(fp, ...)` writes to the stream represented by `fp`.

Remember

Same formatting idea, but a different destination!

Syntax of `fprintf()`

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```
fprintf(fp, format, arguments...)
```

`fp`	The output stream, such as an opened file.
`format`	A string containing ordinary text and format specifiers.
`arguments`	The values to be inserted into the formatted output.

Example

```
int marks = 95;
fprintf(fp, "Marks = %d\n", marks);
```

If `fp` refers to `report.txt`, the file receives:

```
Marks = 95
```

Program: Create a Student Report

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```c
#include <stdio.h>
int main(void)
{
    FILE *fp = fopen("report.txt", "w");
    int roll = 101;
    char name[] = "Asha";
    float marks = 92.5;
    if (fp == NULL)
        return 1;
    fprintf(fp, "Roll: %d\n", roll);
    fprintf(fp, "Name: %s\n", name);
    fprintf(fp, "Marks: %.1f\n", marks);
    fclose(fp);
    return 0;}
```

What does the program do?

1. Opens `report.txt` for writing.
2. Writes the roll number, name, and marks.
3. Closes the file.

What Will Be Stored in the File?

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The program executes these statements:

```
fprintf(fp, "Roll:  %d\n", roll);
fprintf(fp, "Name:  %s\n", name);
fprintf(fp, "Marks:  %.1f\n", marks);
```

The file `report.txt` contains:

```
Roll:  101
Name:  Asha
Marks:  92.5
```

Notice

The output is written to the file, not directly to the console. The program does not need to print anything on the screen.

Understanding the Format Specifiers

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Look at the statement:

```
fprintf(fp, "Marks:  %.1f\n", marks);
```

Part	Purpose
`fp`	Selects the output stream
`Marks:`	Writes ordinary text
`%.1f`	Prints one digit after the decimal point
`\n`	Moves to a new line
`marks`	Supplies the value `92.5`

`92.5` $\longrightarrow$ `Marks:  92.5`

Does `fprintf()` Always Write to a File?

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No! It writes to the stream supplied as its first argument.

Function call	Destination
`printf("Hello");`	Standard output, usually console
`fprintf(stdout, "Hello");`	Standard output, usually console
`fprintf(fp, "Hello");`	Stream associated with `fp`
`fprintf(stderr, "Error!");`	Standard error stream

Key Idea

`fprintf()` can write to the console too. The stream argument determines the destination.

Quick Challenge: Predict the File Contents

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Suppose the program executes:

```
FILE *fp = fopen("output.txt", "w");
int x = 10;
int y = 20;
fprintf(fp, "%d", x);
fprintf(fp, " %d", y);
fclose(fp);
```

What will `output.txt` contain?

1. `10 20`
2. `1020`
3. `x y`
4. Nothing

Hint: Look carefully at the format string in each `fprintf()` call.

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Option 1: 10 20

First statement:

```
fprintf(fp, "%d", x);
```

Writes 10.
Second statement:

```
fprintf(fp, " %d", y);
```

Writes a space followed by 20.

Remember

`fprintf()` does not automatically add spaces or newlines. Include them in the format string when needed.

`fprintf()` vs `printf()`

Both functions can display formatted values. The difference is the **output destination**.

Function	**Where does output go?**
`printf("Hello")`	Standard output, usually the console
`fprintf(fp, "Hello")`	The stream associated with `fp`

Both support format specifiers:

Display on the console

```
printf("Score = %d\n", score);
```

Write to a file

```
fprintf(fp, "Score = %d\n", score);
```

Same formatting power, different output destination!

Trace the Output: Calculate and Save!

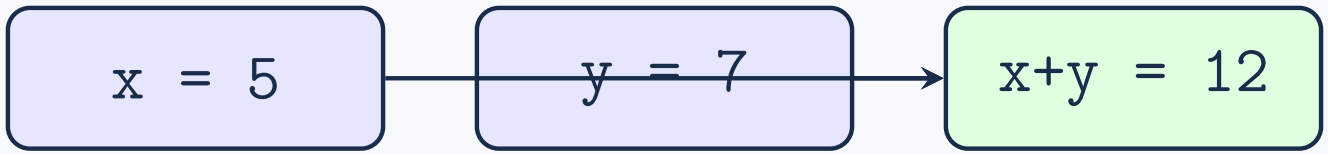
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Suppose:

```
int x = 5, y = 7;
```

Now execute:

```
fprintf(fp, "%d + %d = %d\n", x, y, x + y);
```



The file `result.txt` will contain:

```
5 + 7 = 12
```

Why?

`%d` is replaced by each corresponding integer value, and `\n` adds a newline.

Essential File Functions

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Function	Main purpose	Direction
`fgets()`	Read a line or chunk of text	Stream → Program
`putc()`	Write one character	Program → Stream
`fputs()`	Write a string	Program → Stream
`fscanf()`	Read formatted data	Stream → Program
`fprintf()`	Write formatted data	Program → Stream

A useful distinction

The stream can be a file or another stream, such as `stdout`. In these examples, the write functions are usually used with a file stream.

Choose the Right Function!

What do you need to do?

Read a whole line or chunk of text

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Choose the Right Function!

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What do you need to do?

Read a whole line or chunk of text

`fgets()`

Choose the Right Function!

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What do you need to do?

Read a whole line or chunk of text

`fgets()`

Read an integer and a floating-point value

Choose the Right Function!

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What do you need to do?

Read a whole line or chunk of text

`fgets()`

Read an integer and a floating-point value

`fscanf()`

Choose the Right Function!

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What do you need to do?

Read a whole line or chunk of text

`fgets()`

Read an integer and a floating-point value

`fscanf()`

Write a single character

Choose the Right Function!

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What do you need to do?

Read a whole line or chunk of text

`fgets()`

Read an integer and a floating-point value

`fscanf()`

Write a single character

`putc()`

Choose the Right Function!

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What do you need to do?

Read a whole line or chunk of text

`fgets()`

Read an integer and a floating-point value

`fscanf()`

Write a single character

`putc()`

Write a string without formatting

Choose the Right Function!

॥ सा विद्या या विमुक्तये ॥
भारतीय प्रौद्योगिकी संस्थान धारवाड
In Indian Institute of Technology Dharwad

What do you need to do?

Read a whole line or chunk of text

`fgets()`

Read an integer and a floating-point value

`fscanf()`

Write a single character

`putc()`

Write a string without formatting

`fputs()`

Choose the Right Function!

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Quick Challenge

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Choose the most suitable function for each task.

1. Read a student's name from a text file, one line at a time.
2. Read a roll number and marks using `%d` and `%f`.
3. Write the character `A` to a file.
4. Write `Hello students!` to a file.
5. Save `Total = 125` using a variable and a format specifier.

Think before answering

Are you reading or writing? Do you need formatting?

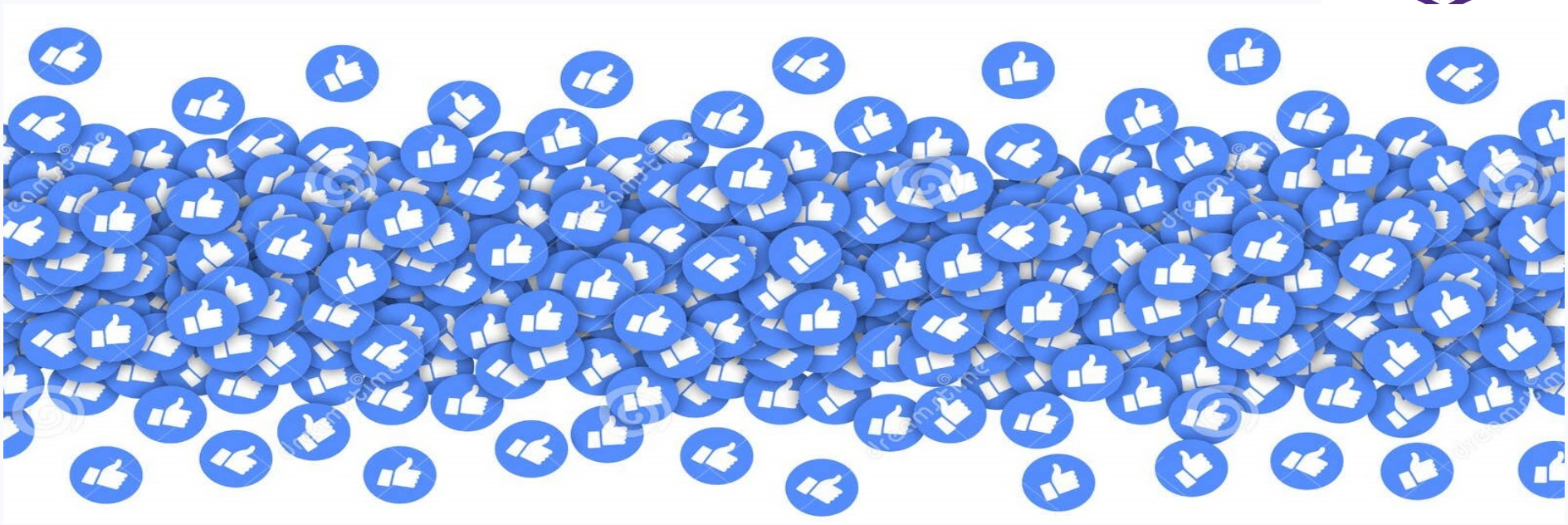
Answers

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Task	Function
1. Read a line	`fgets()`
2. Read formatted values	`fscanf()`
3. Write one character	`putc()`
4. Write a string	`fputs()`
5. Write formatted output	`fprintf()`

Final memory trick

- `fgets`, `fscanf`: read.
- `putc`, `fputs`, `fprintf`: write.



Thank you for your kind attention!!